

Assignment–01

Recurrence Relations

Course Code: CSE 3226
Course Teacher: Md. Imtiage Ahmed
Submission Deadline: 03/07/2026

Instructions

- Solve all questions clearly showing every step.
- Neat presentation and mathematical justification are required.
- Individual hand written submission only.
- Late submission will not be accepted.

Part A: Conceptual Questions

Q1. Define a recurrence relation. Explain why recurrence relations are useful in mathematics and computer science.

Q2. Classify the following recurrence relations according to:

- Order
 - Degree
 - Linear or Non-linear
 - Homogeneous or Non-homogeneous
- (a) $a_n = 2a_{n-1} + 5$
(b) $a_n = 3a_{n-1} - 2a_{n-2}$
(c) $a_n = a_{n-1}a_{n-2} + 1$
(d) $a_n = 4a_{n-1} + 3a_{n-2} + 2^n$
(e) $a_n = \sqrt{a_{n-1}} + a_{n-2}$

Q3. Explain the following with one suitable example each.

- (a) Linear recurrence relation
- (b) Non-linear recurrence relation
- (c) Homogeneous recurrence relation
- (d) Non-homogeneous recurrence relation
- (e) First-order recurrence relation
- (f) Second-order recurrence relation

Part B: Solve Using Substitution (Iteration) Method

Solve the following recurrence relations.

Q4.

$$a_n = a_{n-1} + 3, \quad a_0 = 2$$

Q5.

$$a_n = 2a_{n-1}, \quad a_0 = 5$$

Q6.

$$a_n = 3a_{n-1} + 2, \quad a_0 = 1$$

Q7.

$$a_n = a_{n-1} + 2n, \quad a_0 = 0$$

Part C: Solve Using Characteristic Equation

Find the closed-form solution of the following recurrence relations.

Q8.

$$a_n = 5a_{n-1} - 6a_{n-2}, \quad a_0 = 1, a_1 = 4$$

Q9.

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}, \quad a_0 = 1, a_1 = 2, a_2 = 5$$

Q10.

$$a_n = 6a_{n-1} - 9a_{n-2}, \quad a_0 = 2, a_1 = 6$$

Q11.

$$a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3}, \quad a_0 = 1, a_1 = 3, a_2 = 9$$

Q12.

$$a_n = 2a_{n-1} - 2a_{n-2}, \quad a_0 = 1, a_1 = 2$$

Q13.

$$a_n = -a_{n-2}, \quad a_0 = 1, a_1 = 1$$

Q14. Solve the recurrence relation

$$a_n = 4a_{n-1} - 8a_{n-2} + 8a_{n-3} - 4a_{n-4},$$

given

$$a_0 = 1, a_1 = 2, a_2 = 4, a_3 = 8.$$

Q15. Construct a fourth-order homogeneous recurrence relation whose characteristic roots are

$$1 + i, 1 + i, 1 - i, 1 - i.$$

Then solve the recurrence relation using suitable initial conditions chosen by you.

Analytical Questions

Q16. Find a recurrence relation whose characteristic equation has the roots

$$2, 2, -3, -3.$$

Then obtain its general solution.

Q17. Find a third-order homogeneous recurrence relation whose characteristic roots are

$$2, -1, 3.$$

Then solve the recurrence relation using the initial conditions

$$a_0 = 1, a_1 = 4, a_2 = 11.$$

***** Best of Luck *****